

Supplement to “Benign Exclusion and Consensus-Test Limits in Heterogeneous Federated Learning”

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TABLE XIII
BENIGN UTILITY COST OVER PAIRED DATASET-SEED UNITS
(PERCENTAGE POINTS)

α	Rule	Pairs	Mean	Min.	Max.
0.5	Norm clipping	9	+0.08	−0.07	+0.78
0.5	Coordinate-wise median	9	+3.76	+0.57	+19.07
0.5	Multi-Krum	9	+1.70	−2.71	+10.28
0.5	FLTrust (normalized)	9	+6.02	+1.73	+16.56
0.5	FLAME (HDBSCAN)	9	+7.76	+1.44	+36.85
0.5	Fed-MDBSCAN-G (full)	9	+0.60	−1.22	+6.52
0.1	Norm clipping	9	+0.71	−0.54	+5.40
0.1	Coordinate-wise median	9	+14.37	+4.56	+43.57
0.1	Multi-Krum	9	+12.89	+4.05	+48.49
0.1	FLTrust (normalized)	9	+12.19	+2.77	+33.29
0.1	FLAME (HDBSCAN)	12	+17.66	+6.12	+48.52
0.1	Fed-MDBSCAN-G (full)	12	+6.73	−0.06	+48.42
0.01	Norm clipping	9	+11.92	−2.34	+36.62
0.01	Coordinate-wise median	9	+43.03	+19.61	+67.85
0.01	Multi-Krum	9	+29.83	+2.31	+51.50
0.01	FLTrust (normalized)	9	+9.08	−1.31	+19.19
0.01	FLAME (HDBSCAN)	12	+35.55	+6.92	+68.47
0.01	Fed-MDBSCAN-G (full)	12	+31.60	−0.14	+62.40

Minima and maxima span datasets and seeds, not confidence intervals. The artifact includes each per-seed difference. Pairing checks dataset, partition, initial model and participation schedule hashes. Coverage varies by rule and concentration.

I. PAIRED BENIGN UTILITY MEASUREMENTS

Table XIII summarizes the individual differences used in the adversary-free utility analysis. A pair matches the dataset, concentration, seed, data partition, initial model, and participation schedule. Differences are uniform-mean accuracy minus rule accuracy. The range spans the available dataset-seed pairs; it is neither a confidence interval nor a claim about statistical significance. The artifact file `benign_paired_seed_deltas.csv` retains all individual differences.

II. DUPLICATE-VECTOR DENSITY CHECK

The deployed density function returns 1 if the sum of effective nearest-neighbour distances is exactly zero. This is distinct from the ideal reciprocal-distance expression at strictly positive sums. With $[0, 0, 0.1, 0.2]$ and $k = 1$, it returns $[1, 1, 10, 10]$. Six coincident observations satisfy the zero-sum condition at $k = 5$; a duplicated pair with four distinct neighbours need not. Four small inputs were evaluated in the working source and two frozen sources, giving 12 passing checks. The function ASTs agree. These checks verify the lo-

cal boundary rule, not equivalence to the original MDBSCAN authors’ implementation.

On the recorded update matrices the zero-sum branch is never taken. In all 54 constrained-probe matrices of Section VIII, 18 five-step and 36 canonical, the attacker rows are bit-identical, yet the attackers’ nearest-neighbour distance sums lie between 1.7×10^{-9} and 1.1×10^{-6} . Their relative densities are therefore at least 4.6×10^6 , whereas the largest honest density is 98. At the implementation’s own threshold, the low-density set contains every honest update and no attacker in all 54 matrices. This measurement was not part of the registered plan described in Section IX; it was added after the canonical checkpoint results had been seen.

III. FLAME IDENTITY COUNTS AND SUBSEQUENT INSTRUMENTATION

All 201 canonical FLAME result files match the hashes recorded in the earlier local audit: 200 completed runs and one retained CIFAR failure. The 6,000 completed rounds each contain 90 participants and between 46 and 89 accepted identities. Participant, accepted and rejected identities form disjoint complete partitions, and their confusion counts agree with the exported cell totals. No recorded round uses fallback or accepts everyone. In the frozen selector, the small-cohort and zero-distance skip paths accept everyone, so those paths are excluded by the recorded decisions. Normal selection keeps indices with nonnegative HDBSCAN labels; its minimum cluster size of 46 permits at most one cluster among 90 participants. This establishes the association between saved acceptance-set size and the retained cluster size for these records. It does not reconstruct missing historical labels.

Subsequent instrumentation records whether clustering executed, its skip reason, the configured minimum cluster size, label-derived cluster sizes, the noise count, and the selection size before any fallback. A skipped fit records unknown sizes; an executed all-noise fit records an empty cluster dictionary. The full simulation test suite passes 125 tests. Six bounded frozen/current selector comparisons through the same server preserve all previous non-timing decision fields and produce identical model weights. These checks do not add research training runs or alter the archived results.

IV. CHECKPOINT IDENTITY AND SCOPE

The radius check uses 72 distinct recorded matrices and two gate evaluations per matrix. Alternative gate and cutoff

settings were evaluated offline on the saved matrices and were never fed back into training. Its 36 attacked matrices contain 18 adversarial clients each. All admit the full cohort at the initial stage. The sufficient consensus condition holds in 35, with maximum ratio 2.017225 at the one exception. Gate rows are deduplicated by reference and round before aggregation. With the neighbourhood cutoff enabled, six of the 144 gate evaluations contain candidate groups with members outside the initial pool. All occur outside the attacked block. The subset hypothesis therefore requires an explicit check. Three additional historical batch-regime rows in the supplied radius CSV are not counted as members of the 36-matrix attacked sample. The canonical checkpoint set of Section IX adds 90 matrices, 72 of them attacked.

V. IMPLEMENTATION CHECKS

FLTrust. The local aggregation rule was compared with a transcription of the rule in the authors’ demonstration code on 18 recorded matrices and four root constructions; the relative discrepancy between aggregated updates was below 2.8×10^{-7} . Degenerate-root and zero-trust paths diverge. Root norms were not recorded in the 5,490 canonical rounds, so an unflagged degenerate-root path cannot be excluded in the 135 rounds that accepted every submission. Root and client local-iteration counts also differ, so the check does not establish fidelity of the full training protocol.

FLAME. Six small checks exercise clipping, averaging, and noise scaling through the server’s aggregation path. Equivalence of the HDBSCAN clustering with the authors’ implementation, and the original noise calibration, remain unverified.

Multi-Krum. The local selection was compared with the rule of Blanchard *et al.* on six Fashion-MNIST Min-Max checkpoint matrices of the decision-path replay (seeds 137 and 2024, rounds 0, 9, and 29), through the server’s aggregation path. The server derives $f = 27$, 61 neighbours, and $m = 61$, and its distance computation agrees with an independent float64 computation of the published score to a relative difference of 1.65×10^{-13} . In four of the six matrices the two selected sets have a symmetric difference of four to six members. All differing members belong to a block of 18 byte-identical attacker updates whose tied scores straddle the selection cutoff; the published rule does not specify tie-breaking. The aggregated updates agree to a relative difference below 2.6×10^{-8} in all six matrices, including the two with identical selections. Across the 5,130 canonical Multi-Krum rounds, the recorded cohort is 90, the validity condition $2f + 2 < n$ holds, 61 updates are selected in every round, and no fallback is flagged. The canonical update matrices were not saved, so their scores were not recomputed. The authors’ code was not run, and no multi-round trajectory was compared. The published experiments use $m = n - f$; with $m = 63$ the selected sets would share 59 to 61 of the 61 members.

Cardinality identities. A rule that retained the wrong 61 updates would still have $\text{BER} = 29/90$ in a clean round, so agreement with a cardinality identity is no evidence that its scores or selected identities reproduce the original implementation.

TABLE XIV
DISCRIMINATION BY RULE AND ATTACK FAMILY

Attack family / rule	Cells	TPR (%)	FPR (%)	J (pp)	Benefit (pp)
<i>Loud Gaussian</i> ($\sigma = 5$)					
Multi-Krum	15	100.0	10.4	+89.6	+16.5
FLTrust (normalized)	15	50.0	28.0	+22.0	+14.9
FLAME (HDBSCAN)	15	100.0	31.5	+68.5	+6.2
Fed-MDBSCAN-G (full)	15	100.0	5.8	+94.2	+17.1
<i>Label flipping</i>					
Multi-Krum	9	69.1	22.4	+46.7	−9.5
FLTrust (normalized)	9	79.0	26.7	+52.3	−4.7
FLAME (HDBSCAN)	9	78.5	38.8	+39.7	−14.0
Fed-MDBSCAN-G (full)	9	52.3	11.1	+41.2	−7.9
<i>Patch backdoor</i>					
FLTrust (normalized)	4	31.4	27.6	+3.8	−7.9
FLAME (HDBSCAN)	6	48.4	47.5	+0.9	−26.7
Fed-MDBSCAN-G (full)	6	21.4	20.2	+1.2	−19.2
<i>Bounded random direction</i>					
Multi-Krum	18	33.7	31.7	+2.1	−23.2
FLTrust (normalized)	18	29.7	29.8	−0.1	−9.3
FLAME (HDBSCAN)	18	47.5	46.7	+0.9	−33.0
Fed-MDBSCAN-G (full)	18	25.3	26.7	−1.4	−26.3
<i>Constrained omniscient</i>					
Multi-Krum	6	9.7	41.9	−32.2	−20.1
FLTrust (normalized)	6	66.3	30.7	+35.6	+14.1
FLAME (HDBSCAN)	8	13.1	57.9	−44.7	−19.6
Fed-MDBSCAN-G (full)	8	6.4	20.6	−14.2	−12.9

Means over available cells; columns as in Table VII of the main text. FLTrust counts zero-weight decisions.

Multi-Krum under the constrained probe. In five of the six cells the confusion counts are $\text{TP} = 0$, $\text{FP} = 2,610$, $\text{TN} = 3,060$, and $\text{FN} = 2,430$.

VI. CHECKPOINT DETAILS MOVED FROM THE MAIN TEXT

The median radial ratio over the 36 attacked matrices is 1.448095 and the maximum is 2.017225, at Fashion-MNIST, seed 137, patch, round 29. Promotion to float64 changes Γ by at most 5.43×10^{-7} relative to the native computation and does not alter any threshold classification. Evaluating two gates yields 72 rows from the same 36 matrices, which does not double the sample size. Because the initial and consensus tests use different centres and radii, the convex-hull bound alone does not order their client-level decisions. The 72 replay matrices cover three seeds and rounds 0, 9, and 29. At the clean MNIST checkpoint of the batch-regime example, final rejections are 34, 25, and 18 in regimes A, B, and C; of the 16 clients rejected in A but accepted in C, 10 no longer belong to a tested group and six change at Stage 1. Policy P1 rejects 44 clients in all three regimes, although one pair of identities differs in regime C.

VII. TABLES MOVED FROM THE MAIN TEXT

To keep the main text within its page budget, the per-rule discrimination table and the recovered-cluster example of Section VIII-C appear here. Both are regenerated from the same evidence files as before.

TABLE XV
RECOVERED CLUSTERS AT AN ADVERSARY-FREE MNIST CHECKPOINT

Arm	Cluster	Size	In B_0	Ratio	Decision
A	0	38	38	0.13	accept
A	1	6	6	1.84	reject
A	2	6	0	2.38	reject
A	3	4	4	1.90	reject
A	4	4	0	2.63	reject
A	5	4	4	1.99	reject
B	0	37	37	0.13	accept
B	1	6	6	1.70	reject
B	2	4	4	1.87	reject
C	0	34	34	0.12	accept
C	1	4	4	1.72	reject

Seed 2024, round 9; every client is honest. Mini-batch regimes: A, standard loader with batch size 32; B, a fresh 20-example subset at each local step; C, the first 20-example batch repeated five times. Ratio is $\|c_S - m_{B_0}\|/(\tau R_0)$ with $\tau = 2$ and R_0 as in the consensus test of the main text; a group is rejected when the ratio exceeds 1.

VIII. EXTENDED TABLE NOTES

The notes under the main-text tables give only the definitions needed to read each table. The remaining remarks from earlier versions of those notes are collected here; no result has been removed.

Table III (benign exclusion). Multi-Krum’s BER is $1 - m/n = 29/90 = 32.22\%$ in every cell, independent of dataset and of α (Corollary 1). FLAME’s clean upper bound is $1 - (\lfloor n/2 \rfloor + 1)/n = 44/90 = 48.89\%$ (Corollary 2). CIFAR-10 has no $\alpha = 0.5$ block. FLTrust’s zero-weight count is not a separate rejection stage.

Table IV (benign utility cost). Entries that round to zero are printed without sign. Per-seed paired differences are supplied in the artifact, and Table XIII summarizes their ranges. No significance claim is attached to any single entry.

Table V (FLAME cluster sizes). The frozen normal path retains one non-noise cluster; historical cluster labels were not saved. The minimum size is attained in 719/990 rounds. The counts are not reconstructed from BER.

Table VI (round-level alarm). At $\alpha = 0.01$ every adversary-free round on every dataset raises an alarm. The counts measure false alarms in clean controls, not predictive value at a deployment prevalence. No zero rate is reported for the other rules, because absence of an alarm mechanism is not alarm specificity.

Tables VII and XIV (discrimination). $J > 0$ lies above the chance diagonal. The pooled cells are not independent replicates, and family coverage differs in dataset, concentration and adversary ratio. The summary does not measure radial displacement and is not a controlled displacement experiment. FLTrust has a positive mean J for the constrained family; its counts concern zero weights, and this does not establish a general distinction between defense classes.

Table VIII (constrained adversary). Only the two rules whose admission arithmetic yields a numerical prediction are listed. Multi-Krum discards $n - m = 29$ of 90 submissions every round; in 5 of the six cells it records $\text{TPR} = 0$ with exactly $2,610 = 29 \times 90$ honest rejections, i.e. $\text{FPR} = 29/63 = 46.03\%$, so every discarded submission is honest.

The CIFAR-10/FLAME/ $\alpha = 0.1$ cell also has one retained failure. The conditional identity does not apply to the positive-TPR rows.

Table XI (coverage replay). Widening coverage adds honest rejections and, except in one block, no attacker rejections. That block (P2, density-only gate, patch) reaches $\text{TPR} = 46/324 = 14.20\%$ at $\text{FPR} = 142/1296 = 10.96\%$, i.e. $J = +3.2$ pp; its attack-disabled control records 82 honest rejections under the same policy. Under the density-only gate, P2 would exclude all 1,296 honest Min-Max observations, and the valve restores B_0 in all 18 of those matrices. These are fixed-matrix decision counts; no accuracy or attack success rate is re-measured.

Table XV (recovered clusters). The decision column is the implementation’s own output. The majority cluster lies at 0.12–0.13 of the rejection radius and every honest minority cluster at 1.70–2.63 times it, so the rejection radius separates honest groups from one another.

Table XII (ablation). Removing every layer above the stage-1 acceptance region changes the mean by less than a tenth of a percentage point; no equivalence claim is made from a small mean difference.

IX. TARGETED CHECKS AFTER REVIEW

These checks answer questions raised in an external review of the previous version. Their plan fixed the conditions, measurements, and reporting rules before any outcome run; its SHA-256 digest `cf7dca80...` and time stamp (26 September 2026, 19:08 UTC) are stored with the artifact. The canonical archive was read only. Re-executions used the frozen canonical source and the canonical launch environment, single-threaded and without PyTorch’s deterministic mode. Controls wrap the frozen selection functions at run time instead of modifying the simulation code. No run failed.

Canonical round records. Table XVI counts, for every round of the 201 canonical Fed-MDBSCAN-G runs, whether the gate opened and how many Stage-1-accepted updates the cluster stage removed. At $\alpha = 0.01$ the gate opens in every round of every attack family and in every adversary-free round; at larger α it opens mostly under loud Gaussian noise and label flipping, and on HAR. Under the constrained probes, no Stage-1-accepted update is ever removed. At $\alpha = 0.01$ the adversary-free removals account for 1.0–3.4% of honest rejections on MNIST, Fashion-MNIST, and CIFAR-10, and for 28.3% on HAR.

Canonical checkpoint matrices. Thirty canonical Fed-MDBSCAN-G runs were re-executed and captured at rounds 0, 9, and 29 (Table XVII); its two group-count columns were added after the results had been seen. All 30 reproduced every non-timing field of their 30 archived round records. Under the constrained probes, B_0 contains every attacker in all 36 matrices, and the forced cluster stage recovers no candidate group: the copied attackers form the high-density side of the partition, and no group emerges among the honest updates that are tested. The sufficient condition therefore holds vacuously for the groups, and $\Gamma \leq \tau$ in 13 of these 36 matrices. Under the patch attack at $\alpha = 0.1$, recovered groups exist in all 18 matrices and lie inside B_0 , but none is rejected. In the patch

TABLE XVI
CLUSTER-STAGE ACTIVITY IN THE CANONICAL FED-MDBSCAN-G
RUNS

Attack family	α	Rounds	Gate open	Rounds removing from B_0	Removed updates
Adversary-free	0.5	270	19	0	0
	0.1	360	89	26	200
	0.01	360	360	67	738
Constrained probes	0.1	360	101	0	0
	0.01	360	360	0	0
Patch backdoor	0.1	270	0	0	0
	0.01	270	270	89	397
Loud Gaussian	0.5	270	270	0	0
	0.1	540	540	24	114
	0.01	540	540	151	777
Label flipping	0.5	270	241	18	32
	0.1	270	201	25	117
	0.01	270	270	94	825
Bounded perturbation	0.1	540	177	60	444
	0.01	1080	1080	266	2978

Counts over every dataset and seed of the canonical main block, read from the recorded round fields; no run was repeated. The gate is open when the recorded layer is not Stage 1 alone. A removed update is a Stage-1-accepted update that the cluster stage excludes from the final set, $|B_0| - |B|$; in adversary-free rounds all of them are honest.

and adversary-free matrices at $\alpha = 0.01$, Γ lies far above τ , with medians 5.5 and 5.7. Stage 1 already excludes 190 of the 486 patch attackers there, and the test removes Stage-1-accepted updates in 8 of these 36 matrices.

FLAME clustering against the reference library. Nine canonical FLAME runs were re-executed: MNIST at $\alpha = 0.5$ and 0.01 and HAR at $\alpha = 0.01$, three seeds each (Table XVIII). All nine reproduced their archived round records. In each of the 270 rounds, the reference `hdbscan` library with `min_samples = 1` selected exactly the updates that the local `scikit-learn` selector with `min_samples = 2` selected. This covers the clustering step on these inputs; the noise calibration and the authors’ full training protocol remain unverified.

Noise and random-selection controls. Tables XIX and XX report the adversary-free controls at $\alpha = 0.01$. Setting FLAME’s noise to zero changes final accuracy by at most 0.33 points. Replacing Multi-Krum’s 61 selected updates with a random subset of the same size raises accuracy by 15.9 to 43.7 points on average, to within 1.3 points of uniform averaging on MNIST and Fashion-MNIST and 4.9 points above it on HAR. For FLAME, the same replacement raises accuracy by 6.6 to 25.4 points, but a gap of 8.6 to 40.1 points to uniform averaging remains. The control attributes that gap to FLAME’s clipping, its smaller admitted set, or both, and does not separate the two. The random subsets overlap the rules’ selections at chance level: 68% for Multi-Krum and 52–60% for FLAME.

X. REPRODUCTION BOUNDARY

From the V5 submission directory, run `python analysis/build_tables.py --root .` with the existing project Python environment. It checks the copied report hashes and condition metadata, regenerates the tables and figures, and verifies the direct-count and checkpoint

summaries. The main and supplement are compiled with the supplied IEEEtran files. A repository clone suffices for these summary-level checks. Repeating the historical identity-level audit or recapturing training checkpoints requires the separate local raw archive and frozen environment. Hash equality establishes content integrity, not a hypothesis’s timing or preregistration.

The five failed canonical units remain failures. Deterministic CIFAR diagnostics are not replacements for the failed unit. No new backdoor improvement, statistical equivalence, or general defense ranking is inferred from the analyses in this supplement.

TABLE XVII
DECISION-PATH MEASUREMENTS ON CANONICAL CHECKPOINT MATRICES

Condition	Exact runs	Matrices	Median Γ	$\Gamma \leq \tau$	With groups	Groups in B_0	Sufficient condition	Deployed gate ran	Matrices with removal	Removed honest/att.	Attackers outside B_0
Min-Max, $\alpha = 0.1$	6/6	18	1.96	10	0	–	10	0	0	0/0	0/486
Min-Sum, $\alpha = 0.01$	6/6	18	2.43	3	0	–	3	18	0	0/0	0/486
Patch, $\alpha = 0.1$	6/6	18	2.02	9	18	18	9	0	0	0/0	2/486
Patch, $\alpha = 0.01$	6/6	18	5.52	0	17	13	0	18	7	30/13	190/486
Adversary-free, $\alpha = 0.01$	6/6	18	5.73	0	18	13	0	18	1	4/0	–

Canonical Fed-MDBSCAN-G runs on MNIST and Fashion-MNIST, three seeds each, re-executed with the frozen canonical source; “Exact” counts runs whose 30 recorded rounds matched the archive in every non-timing field. Matrices are the server inputs of rounds 0, 9, and 29. “With groups” counts matrices on which the cluster stage, forced to run by the density-only gate, recovers at least one candidate group; “Groups in B_0 ” counts those on which every recovered group lies in B_0 . “Sufficient condition” counts matrices with $\Gamma \leq \tau = 2$ on which every recovered group, if any, lies in B_0 . Removal columns refer to the deployed gate. Attackers outside B_0 were excluded by Stage 1.

TABLE XVIII
FLAME CLUSTERING: LOCAL SELECTOR VERSUS REFERENCE
HDBSCAN

Clean cell	Exact runs	Clustered rounds	Identical selections	Max. sym. difference
MNIST, $\alpha = 0.5$	3/3	90	90	0
MNIST, $\alpha = 0.01$	3/3	90	90	0
HAR, $\alpha = 0.01$	3/3	90	90	0

Canonical FLAME runs were re-executed; in every round the precomputed cosine-distance matrix of the local selector was also passed to the reference hdbscan library (0.8.44) with `min_cluster_size =  $\lfloor n/2 \rfloor + 1$` , `min_samples = 1`, and a single cluster allowed. The runs kept the local selection, so reproduction of the archive could be checked.

TABLE XIX
ADVERSARY-FREE FINAL ACCURACY AT $\alpha = 0.01$ WITH NOISE AND SELECTION CONTROLS

Dataset	FedAvg	FLAME	FLAME no noise	FLAME random subset	Multi-Krum	Multi-Krum random subset
MNIST	85.65	20.10	20.08	45.53	40.71	84.36
Fashion-MNIST	73.63	27.74	27.61	43.79	40.04	72.58
UCI HAR	45.53	30.35	30.37	36.93	34.57	50.44

Mean final accuracy (%) over seeds 42, 137, and 2024. FedAvg, FLAME, and Multi-Krum are the canonical runs. “No noise” sets the FLAME noise factor to zero. “Random subset” replaces the selected updates in every round with a uniformly random subset of the same size, so clipping, noise, and the number of admitted updates are unchanged. All controls share each seed’s data partition, participation schedule, and initial model with the canonical runs.

TABLE XX
PAIRED FINAL-ACCURACY DIFFERENCES FOR THE CONTROLS OF TABLE XIX

Dataset	No noise – FLAME	FLAME – random subset	Multi-Krum – random subset	FedAvg – Multi-Krum random
MNIST	−0.01 [−0.02, +0.00]	−25.44 [−35.97, −8.60]	−43.65 [−49.69, −35.88]	+1.29 [−0.86, +2.92]
Fashion-MNIST	−0.12 [−0.33, +0.00]	−16.06 [−21.97, −4.73]	−32.54 [−37.07, −30.23]	+1.05 [−0.43, +2.36]
UCI HAR	+0.02 [+0.00, +0.07]	−6.58 [−13.67, +3.97]	−15.87 [−27.32, −2.41]	−4.91 [−11.33, −0.10]

Percentage points, mean [minimum, maximum] over the three seed pairs. A negative entry in the FLAME or Multi-Krum column means that the rule’s own selection yields lower accuracy than a random subset of the same size.